The ReACT Agent Loop (Per-Turn Reason + Act + Observe)

> **Scope.** This document describes the **core ReACT agent loop** — the per-turn loop that runs every time you send a message to ragent. It is *not* the goal-driven `/loop` feature (see `TUI-QUICKSTART.md`, section 5, for `/loop`). Where the two interact, the interaction is called out explicitly.

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1. What it is

Every user message triggers an agentic **Reason -> Act -> Observe** cycle:

1. **Reason** — the session processor streams an LLM response for the current conversation state.
2. **Act** — if the model requests tool calls, each call is permission-checked and executed.
3. **Observe** — the tool results are appended to the conversation history and fed back into the next LLM request.

The cycle repeats until the model produces an answer **without** any tool calls, or a termination condition fires (step limit, cancel, interrupt, or an unrecoverable error).

Element	Location
Orchestrator struct	`SessionProcessor` — `crates/ragent-agent/src/session/processor.rs:106`
Public entry (plain text prompt)	`process_message` — `processor.rs:1137`
Public entry (pre-built `Message`, multipart/images)	`process_user_message` — `processor.rs:1164`
The core loop	inline `loop { ... }` — `processor.rs:1472`, inside `process_user_message`
Per-step helpers	`crates/ragent-agent/src/session/loop_steps.rs` — `prepare_client` (:121), `build_turn_system_prompt` (:362), `build_turn_chat_messages` (:649), `run_inline_init_acknowledgement` (:739), `call_llm_step` (:857), `finalize_assistant_message` (:1529)
Loop-run state machine	`LoopSpec` / `LoopTracker` / `StopCondition` — `crates/ragent-agent/src/session/loop_state.rs`

`SessionProcessor` orchestrates the loop: it accepts a user message, streams an LLM response, executes any requested tool calls, and iterates until the model signals completion or the step limit is reached.

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2. Turn lifecycle

Phase A — turn setup (before the loop)

Location: `processor.rs:1171-1465`.

1. **Telemetry** — an `agent_loop_profiler()` and a `SessionRecorder` are created and `record_session_start()` is logged.
2. **User message persisted** — `storage_op(move |s| s.create_message(&msg))` writes the message to SQLite on a blocking thread; `Event::MessageStart` is published.
3. **RunId issued** — `RunId::new()`; the message is recorded in the activity log as a user model message. This `run_id` links every event of the turn.
4. **Client/model/config resolution** — `prepare_client` resolves the provider client, model reference, session config, working directory and team context. Failure here is an unrecoverable provider-stage error: the run is terminated with status `error` and `Err` is returned.
5. **Per-run cost tracking** — a `TokenUsage` listener accumulates `(input, output)` tokens for the turn; when the turn ends a `RunCostSummary` row is computed from merged price entries and published as `Event::RunCostSummary`.
6. **System prompt build** — `build_turn_system_prompt` assembles the agent preset, initiatives, tool reference, codeindex guidance and team guidance via the cached `SystemPromptCache`.
7. **Chat history build** — `build_turn_chat_messages` returns the provider chat messages, whether compression happened, the last reported input token count, and the context window. The vector is shared behind an `Arc<Vec<ChatMessage>>` so per-retry requests share it by refcount.
8. **AGENTS.md init acknowledgement** — `run_inline_init_acknowledgement` prints a one-time display-only acknowledgement (skipped for sub-agents).
9. **Loop setup**:
   - `max_steps = agent.max_steps.unwrap_or(1024)` — the default iteration ceiling is **1024** unless the agent preset sets one.
   - `event_bus.set_step(session_id, 0)`.
   - Tool definitions are chosen: **empty if `max_steps <= 1`** (single-shot: the model gets no tools); filtered to the active `/loop` tool set when one is active; otherwise the full cached tool registry.
   - A reusable assistant placeholder `Message` is created and persisted — its id anchors all streaming updates to one DB row.
   - Interrupt plumbing: `interrupt_requested` = user cancel flag OR active `/loop` interrupt flag.

Phase B — one ReACT iteration

Each pass of the `loop {` body (`processor.rs:1472+`) runs this guard chain in order:

#	Guard	Behaviour
1	Stop-flag guard	If the loop tracker `is_stopped()`, break — no stage runs.
2	Safe-point interrupt	On `/loop` runs, a user cancel **or** a raised `/loop` interrupt breaks **before** the budget gate (first stop wins; user intervention takes precedence) and ends the run with status `interrupted`.

#	Guard	Behaviour
3	Budget gate	`tracker.begin_step()` returns false when the `/loop` step or token budget was reached; the run ends as `budget_exhausted` **before another LLM request is sent**.
4	Step counter	`set_step(current + 1)` under the `loop.step.total` profile scope.
5	Max-steps check	`step > max_steps` -> warning + `Event::AgentError("Reached maximum steps (N)")` + break.
6	User-cancel check	On plain (non-loop) turns, a set cancel flag logs the timing breakdown, persists the partial assistant message, publishes `Event::MessageEnd { reason: Cancelled }`, records `TerminationReason::Interrupted`, publishes the run-cost summary, and returns an empty message. On `/loop` runs this branch is effectively plain-turn-only — a loop-run cancel already ended the run at guard 2.

Then the reasoning stage:

7. `Event::ToolsSent` — published **only on step 1 and only when tools are offered** (`!tool_definitions.is_empty()`), so the TUI renders the offered tool list once.
8. **Pre-send auto-compaction** — when `compaction.auto` is enabled and not yet attempted this turn, the request is token-estimated and `evaluate_trigger` runs against the context window; if the trigger fires, history is summarised (`crate::compaction::compact`), a synthetic `Role::Compaction` message is persisted, and the in-memory history is swapped to compaction-summary + verbatim recent tail. Compaction failure only warns and continues uncompressed.
9. **LLM call** — `call_llm_step` builds the `ChatRequest` (system prompt, history, tool definitions) and folds the provider stream (`futures::StreamExt`) into incremental `text_buffer`, `reasoning_buffer` and accumulated `tool_calls`. A fatal stream error terminates the run with status `error`.
10. **Provider-reported input tokens** — persisted into session state so the TUI context panel shows the same figure next turn.
11. **Token tally** — `tracker.record_tokens(...)` is persisted; the *next* iteration's budget gate sees the updated totals.
12. **Response parts** — non-empty reasoning becomes a `MessagePart::Reasoning` on the assistant

message; non-empty text publishes `Event::ModelResponse` (preview, elapsed ms, token counts) and becomes a `MessagePart::Text`.

Phase C — decision point: tool call vs final answer

```
if llm_result.tool_calls.is_empty()
    -> the text response IS the final answer; the loop ends
else
    -> tool dispatch phase; afterwards `continue` back to Phase B
```

When the model answers **without** tool calls, three sub-paths run (in order):

- **Sub-agent summary nudge** — if this is a sub-agent run past step 1 whose text is under the 2000-byte narration limit, the narration was likely "Now let me check…" rather than findings. The narration is popped, pushed back into history as an assistant message, and a `SUBAGENT_SUMMARY_NUDGE` user message requests the real summary; the loop continues. This happens **once per run** and is *not* a termination.
- **Verification gate** (`/loop` only, T-005) — when a spec with a `verify_cmd` is active, `run_verification_command` decides: **pass** -> the run ends as `GoalAchieved` with the verify label; **fail with steps remaining** -> the failure output becomes the next observation and the loop continues (a success record resets the consecutive-failure counter so unrelated recoverable failures do not burn the retry allowance); **fail with budget breached** -> ends as `budget_exhausted`.
- **Plain stop** — a no-tool-call response with no verification command means the goal was reached -> run ends as `GoalAchieved`.

When tool calls are **non-empty**, one more interrupt safe point runs, then the **tool dispatch phase** (`processor.rs:2042+`): text is moved (not cloned) into a `ContentPart::Text`, `parallel_tool_calls` is read from `experimental.parallel_tool_calls`, one `ToolContext` is built per step, and per tool call: **permission check -> execution -> result collection -> interim save**, then `continue` back to the top of the loop so the next LLM request carries the observations.

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3. Flow diagram

```
User message
     |
     v
persist message -> RunId issued -> prepare_client (fatal on error)
     |
     v
build system prompt + chat history (cached)
     |
     +-------------------------------------------------+
     |  loop {  (processor.rs:1472)                    |
     |    stop-flag? -> break                          |
     |    /loop interrupt (or cancel on loop runs)?    |
     |      -> status interrupted, break               |
     |    budget breach? -> budget_exhausted, break    |
     |    step > max_steps? -> AgentError, break       |
```

```
|    plain-turn cancel? -> persist partial,        |
|      MessageEnd{Cancelled}                       |
|    ToolsSent (step 1 only)                       |
|    auto-compaction (if enabled, once per turn)   |
|    call_llm_step  <-- REASON (stream fold)       |
|    record tokens / response parts                |
|                                                  |
|    tool_calls empty?  --yes--> final answer, end |
|        |no                                       |
|        v                                         |
|    interrupt safe point                          |
|    per tool: permission -> execute -> observe    |
|    append tool results -> continue loop          |
+--------------------------------------------------+
```

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4. Tool calls vs final answer

The decision is purely structural: a streamed response that accumulates one or more `tool_calls` is an **action** turn; a response with **no** tool calls is the **final answer** and terminates the loop. Thinking/reasoning blocks are buffered separately (`reasoning_buffer`) and persisted as `MessagePart::Reasoning` — they never terminate the loop on their own.

Tool-call arguments arrive incrementally; assembly is managed through `PendingToolCall` (history.rs) before a call is promoted into `llm_result.tool_calls` and dispatched.

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5. Permissions and tool execution

Tool execution is gated by the multi-layered permission system:

- `PermissionChecker` consults global rules (`permission` array in `ragent.json`) and per-agent rules (`agent.<name>.permission`).
- Allow/Deny verdicts are honoured; an Allow that would auto-approve can still be forced into an interactive prompt when a `/loop` spec with `checkpoints: true` is active (destructive tools are checkpointed).
- `--yes` / `--no-prompt` (CLI) or `/yolo` (TUI) auto-approve every prompt; explicit Deny rules still take precedence over allow rules.
- A permission prompt times out to **deny** (safe default).

A tool task that neither finishes nor panics is killed at the tool watchdog timeout (`TOOL_WATCHDOG_TIMEOUT` = 5000 s) and the whole run is terminated with status `error`; an ordinary task join failure is logged and skipped instead.

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6. Streaming, stalls, retries

- **Stream driver**: `call_llm_step` folds `StreamEvents` into the per-step working state (`LoopState`): incremental text, reasoning, tool calls.

- **Thinking blocks** are configured per provider/model (see Section 8, `thinking`) and persisted as `MessagePart::Reasoning`.
- **Stall detection** — `stream.timeout_secs` bounds the gap between subsequent stream deltas; a stall triggers the retry path. Stalls are detected via `stall_pattern_set()` with `should_flush`.
- **Retry/backoff** — `stream.max_retries` attempts with linear backoff (`N * stream.retry_backoff_secs`). Retryability helpers: `should_retry_stream_error`, `stream_has_meaningful_partial_output`, `is_token_overflow_error_message`, `is_permanent_llm_api_error`.
- **Initial response timeout** — `stream.initial_response_timeout_secs` waits for the first byte (RTT + provider cold start).

UI event fan-out during a turn: `MessageStart` -> step counter -> `ToolsSent` (step 1) -> `ModelResponse` -> per-tool `ToolCallArgs` / `ToolCallEnd` / `ToolCallStart` (inside the spawned tool task; `ToolCallBatch` at step end) -> `MessageEnd`.

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7. Termination conditions

#	Condition	Signal / status
1	Model answers without tool calls	`GoalAchieved` (plain turns simply return the message)
2	`/loop` verification gate passes	`GoalAchieved` with the verify label
3	Step budget exhausted	`AgentError "Reached maximum steps (N)"`; default ceiling 1024 (agent override) / `--maxsteps` cap
4	`/loop` token cost budget exhausted	`BudgetExhausted`, fired before the next request
5	User cancel	`FinishReason::Cancelled`, partial message persisted
6	Human interrupt (`Esc` during a `/loop` run)	`HumanIntervention` -> status `interrupted`; takes precedence over budget at safe points
7	Unrecoverable provider error (prep or stream stage)	status `error`, `Err` returned
8	Tool watchdog	tool stalls past 5000 s -> run terminated

Not terminations: the one-shot sub-agent summary nudge, and verification failures that still have steps remaining (they become the next observation).

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8. Configuration options

ragent reads configuration from `ragent.json` (or `ragent.jsonc`) with the following precedence, each later layer merged over the previous:

1. Compiled defaults
2. Global file: `~/.config/ragent/ragent.json`
3. Project file: `./.ragent/ragent.json` (created with defaults if neither file exists)
4. `RAGENT_CONFIG` env var -> file path overlay
5. `RAGENT_CONFIG_CONTENT` env var -> inline JSON overlay

Per-provider blocks and per-model entries are deep-merged, so partial overrides preserve lower layers. Resolved config is cached keyed on the mtime + size of every contributing file plus the cwd (`CachedConfigFile`); every `Config::save()` invalidates that cache so same-process writes are immediately visible to subsequent loads.

Core loop knobs

Key	Type	Default	Behaviour
`defaultAgent`	`String`	`"general"`	Default agent preset driving the loop.
`agent.<name>.max_steps`	`Option<u32>`	unset (falls back to **1024**)	Iteration ceiling for the loop when this agent is active.
`agent.<name>.model`	`String`	unset	`provider:model` override for this agent.
`agent.<name>.temperature`	`f32`	unset	Sampling temperature.
`agent.<name>.top_p`	`f32`	unset	Nucleus sampling.
`agent.<name>.permission`	rules	`[]`	Per-agent permission rules.
`permission`	`Vec<PermissionRule>`	`[]`	Global allow/deny/ask rules (`permission`, `pattern`, `action`).
`yolo`	`bool`	`false`	Trust mode: auto-approve everything.
`experimental.parallel_tool_calls`	`bool`	`false`	Execute requested tool calls in parallel rather than sequentially.
`experimental.max_background_agents`	`usize`	`8`	Concurrent background sub-agent cap.
`experimental.background_agent_timeout`	`u64`	`3600`	Sub-agent timeout (seconds).

`agent_perf` — per-step performance tuning

Consulted at loop startup and on every step. `validate()` refuses to start on violations.

Key	Type	Default	Behaviour
`agent_perf.enabled`	`bool`	`true`	Master switch; `false` short-circuits every perf optimisation.

Key	Type	Default	Behaviour
`agent_perf.profiling`	`bool`	`false`	Per-scope timing logs at `info` level.
`agent_perf.step_budget_secs`	`u64`	`300`	Max wall-clock seconds per agent step (>= 5).
`agent_perf.stall_timeout_secs`	`u64`	`60`	Seconds without a stream delta before stall recovery (>= 5).
`agent_perf.max_concurrent_tools`	`u32`	`min(available_parallelism, 4)`	Max parallel tool calls per turn (>= 1).
`agent_perf.parallel_independent_tools`	`bool`	`true`	Execute independent tool calls in parallel.

`stream` — request timeouts and retries

Key	Type	Default	Behaviour
`stream.initial_response_timeout_secs`	`u64`	`300`	Wait for the **first byte**.
`stream.timeout_secs`	`u64`	`120`	Max gap between **subsequent** deltas before stall -> retry.
`stream.max_retries`	`u32`	`4`	Retry attempts after stall/connection failure (<= 32).
`stream.retry_backoff_secs`	`u64`	`2`	Attempt N waits `N * retry_backoff_secs`.

Validation: both timeouts >= 5; `initial_response_timeout_secs >= timeout_secs`.

`compaction` — history compaction

Key	Type	Default	Behaviour
`compaction.auto`	`bool`	`true`	Auto-compact before a send when the trigger fires (once per turn).
`compaction.threshold`	`f32`	`0.7`	Fraction of context window that triggers compaction.
`compaction.buffer`	`f32`	`0.10`	Headroom kept after compaction.
`compaction.keep.tokens`	`f32`	`0.20`	Fraction of recent tokens kept verbatim after compaction.

`thinking` — reasoning blocks

Configured per provider and per model; model-level overrides provider-level.

Key	Type	Default	Behaviour
`provider.<id>.thinking.enabled`	bool	provider-dependent	Enable thinking blocks for the provider.
`provider.<id>.thinking.level`	String	provider-dependent	Reasoning effort (e.g. `low`/`medium`/`high`).
`provider.<id>.models.<mid>.thinking.budget_tokens`	u32	provider-dependent	Token budget for reasoning.

Tool surface

Every tool the registry holds is advertised to the model: the definitions sent with each request are the registry's definitions, and the system prompt lists the same set. There is no switch that hides a family from the model; a missing tool means the subsystem that registers it is not active (for example the code index registers its ten tools only while `/codeindex on`). `/tools list` prints the live set.

The `loop` section — `/loop` only, not the core loop

The `loop` section (`loop.max_steps`, `loop.cost_limit`, `loop.error_retry_allowance`, `loop.checkpoints`, `loop.checkpoint_timeout_secs`) configures the **goal-driven `/loop` feature**, which injects stop conditions and a verification gate into the core loop at its safe points. It does not change how a plain per-turn turn runs.

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9. Commands

CLI (`src/main.rs`)

Flag / command	Effect
`--model <MODEL>`	Override provider/model for the loop.
`--agent <AGENT>`	Agent preset driving the loop (default `general`).
`--yes` (alias `--no-prompt`)	Auto-approve all permission prompts.
`--no-tui`	Headless stdout instead of the TUI.
`--maxsteps <N>`	Cap agentic loop steps (CLI help text says 500, but with no agent override the effective ceiling is the agent preset's `max_steps` — default 1024).
`--no-git-context` / `--no-readme-context`	Disable git/README context injection into the system prompt.
`--dry-run`	Readiness check only; never invokes model or tool.
`ragent run <PROMPT>`	One-shot agent execution of a single prompt.
`ragent serve [--addr <ADDR>]`	Start the HTTP server (default `127.0.0.1:3000`).
`ragent session resume <ID>`	Resume a session into a live loop.

TUI slash commands

Command	Effect on the loop
`/agent [<name>]`	Switch the agent preset driving the loop.
`/model [<id>]`	Switch model mid-session.
`/thinking`	Configure thinking mode.
`/tools`	Tool registry / visibility control.
`/compact`	Force context compaction of loop history.
`/cost`	Show the run-cost summary.
`/cancel`	Cancel in-flight processing.
`/clear`	Clear the session transcript.
`/system`	Show the current system prompt.
`/undo`	Roll back to the last file snapshot.
`/yolo`	Toggle YOLO trust mode (auto-approve).
`/autopilot on [--max-tokens N] [--max-time N]` / `off` / `status`	Autonomous loop continuation with token/time budget.
`/reload [agents|config|mcp|skills|all]`	Hot-reload subsystems into the running loop.
`/context refresh`	Clear the prompt-context cache (file tree/git/README recomputed next turn).

HTTP API (Bearer token required; `GET /health` is open)

Endpoint	Shape
`POST /sessions/{id}/messages`	Body `{"content": String}` — the primary loop-driving endpoint.
`POST /sessions/{id}/abort`	Aborts the running loop.
`POST /sessions/{id}/permission/{req_id}`	Body `{"decision": "allow"|"always"|"deny"}`.
`GET /events`	SSE event stream (`MessageStart`, `ModelResponse`, `ToolCallEnd`, …).
`POST /sessions/{id}/tasks`	Spawn a sub-agent (`agent`, `task`, `background?`, `model?`).

Keyboard shortcuts

- **Esc** during processing — request a loop interrupt (“stopping after the current stage”).
- **Ctrl+C** — copy-to-clipboard when text selection is active, else quit arm; **Ctrl+D** confirms quit.

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10. Worked examples

One-shot CLI run

```
export ANTHROPIC_API_KEY="sk-..."
ragent run --agent coder "Refactor src/main.rs to split the CLI wiring"
```

Driving the loop over HTTP

```
# create a session
SID=$(curl -s -X POST http://127.0.0.1:3000/sessions \
  -H "Authorization: Bearer $RAGENT_TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"directory": "."}' | jq -r .id)

# drive the loop
curl -s -X POST http://localhost:3000/sessions/$SID/messages \
  -H "Authorization: Bearer $RAGENT_TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"content": "List the files in crates/ and summarise each crate"}'

# watch events
curl -N http://localhost:3000/events -H "Authorization: Bearer $RAGENT_TOKEN"
```

Autopilot continuation

```
/autopilot on --max-tokens 50000 --max-time 600
```

The loop continues autonomously until the task completes, the budgets are exhausted, or `/autopilot off` is issued.

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11. Observability

- **Run cost** — every turn publishes `Event::RunCostSummary` with accumulated input/output tokens and cost derived from merged `prices` entries (`PriceEntry { model, input_per_1m, output_per_1m }`, USD per 1M tokens).
- **Activity log** — append-only JSONL event store keyed by `RunId`; every user/assistant message and termination reason is recorded.
- **LLM request stats** — `llmstats` slash command shows request statistics (counts, durations, token tallies).
- **OpenTelemetry** — `telemetry` section (or legacy `experimental.openTelemetry`) enables OTLP export of session spans.

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12. Related documentation

- `TUI-QUICKSTART.md` — using the terminal UI (section 5 covers `/loop`)
- `QUICKSTART.md` — installation and first-run configuration
- `SPEC.md` — full configuration schema
- `docs/howtos/loopprogramming.md` — the goal-driven `/loop` extension that wraps this loop with a spec and stop conditions
- `docs/howtos/office.md` — the PDF tool family this loop can execute
- `docs/howtos/teams.md` — multi-agent team coordination
- `docs/performance/benchmark-guide.md` — benchmarking the loop surfaces